

# Treasury Forecast Tracker

A Microsoft Fabric and Power BI demonstration | Brett McDonald | 8 October 2026

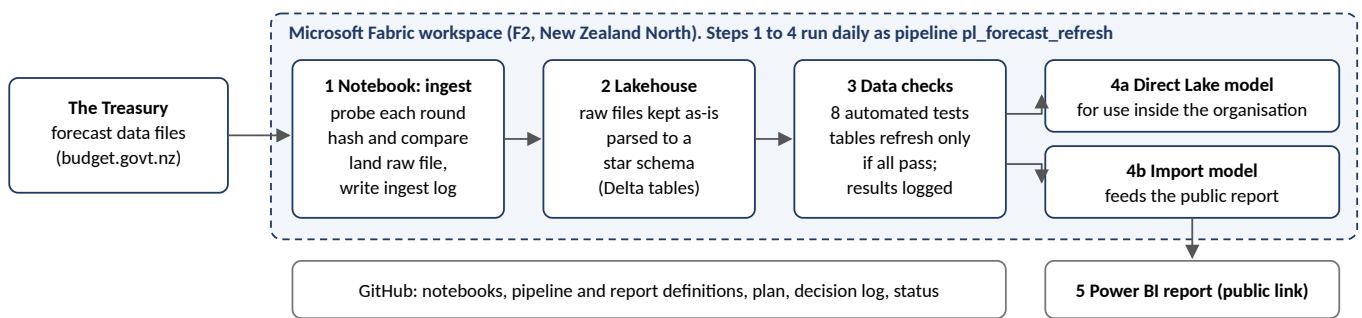
Live report (no sign-in): [open the Power BI report](#) | Code and project documents: [github.com/brett-mcdonald-projects/treasury-forecast-tracker](#)

## What it does

The report answers one question: **how has the economic and fiscal outlook changed across the Treasury's successive forecast rounds?** It loads the Treasury's published Economic and Fiscal Update data files for three rounds (Half Year Update 2025, Budget Update 2026 and Pre-election Update 2026) and lets the reader compare any of eight headline measures across them, such as real GDP growth, CPI inflation, the operating balance and net core Crown debt.

I built it to show how I approach a reporting solution on Microsoft Fabric end to end: automated ingestion, a tested data model, and a governed route from source file to published report. It is an independent demonstration on public data and is not a Treasury product.

## How it works



## What is built and working today

| Component             | Detail                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source registry       | One row per forecast round with its file address. A new round or a new source is a new row, not a redesign. Four rounds are registered. The fourth, Half Year Update 2026, has no file yet; the Treasury has said the timing of its next forecast update will be determined later.                                                                                                                               |
| Change detection      | Each run requests every registered file, hashes it (SHA-256) and compares it with the last loaded hash in the ingest log. Only new or changed files are landed and processed. A missing file is recorded, not treated as a failure.                                                                                                                                                                              |
| Lakehouse             | Raw files are kept untouched in a timestamped landing folder. Parsed data is held in eight Delta tables: 12,411 observations across 48 Treasury series and three rounds, 1,421 rows of derived headline measures, plus the registry, ingest log and test results.                                                                                                                                                |
| Data checks           | Eight tests run before any table is refreshed: row presence, duplicate keys, nulls, referential integrity, required series present in every round, no gaps in the quarterly GDP series, forecast flag complete, and a plausibility range. A failed test stops the refresh.                                                                                                                                       |
| Pipeline and schedule | A Fabric pipeline, pl_forecast_refresh, runs two steps in order every day at 7:00 am: the ingest notebook, then a second notebook that syncs the lakehouse SQL endpoint, refreshes the public report's model and waits for the result. The second step runs only if the first succeeds, and a failed refresh fails the run.                                                                                      |
| Semantic models       | A Direct Lake model over the lakehouse, and an import-mode copy for the public report, because Power BI's public publishing does not support Direct Lake. The import model carries a small set of DAX measures: the insight sentence, the unit labels, and a number format that follows the selected measure.                                                                                                    |
| Report                | Two pages, linked by buttons. <i>Forecast tracker</i> : measure selector, a sentence calculated from the data that states what the latest round changed for the selected measure, and a chart and table by round, each labelled with its unit of measure. <i>Pipeline and data quality</i> : every run of the ingest step with what it found, and the test results, so the reader can see what the pipeline did. |

**Runs on 8 October 2026:** the first load (10:48 am) fetched three rounds directly from [budget.govt.nz](https://budget.govt.nz) (HTTP 200) and passed 8 of 8 data checks; Half Year Update 2026 returned 404 and was logged as not published. The pipeline then ran end to end twice (1:49 pm and 1:53 pm, each in under three minutes). Both runs found the three files unchanged by hash, so nothing was reloaded. The first run refreshed the model seconds before the lakehouse SQL endpoint had picked up the new log rows; a sync step was added and the second run confirmed the fix. The first scheduled run is due at 7:00 am on 9 October.

## Design choices

- **Definitions are computed once, in the data layer.** Growth rates and percent-of-GDP measures are derived in the notebook, so every report built on the model uses the same calculation.
- **Raw data is never overwritten.** Each fetched file is stored as received, so any figure can be traced back to the file it came from.
- **The pipeline reports on itself.** The ingest log and test results are tables in the model and are shown in the report.
- **The report is held as code.** The layout and the measures were applied through the Fabric REST API, and the report, pipeline and measure definitions are stored in the repository alongside the notebooks.
- **Neutral presentation.** The report describes how forecasts moved between rounds. It does not assess forecast quality.
- **One documented heuristic.** The economic data file has no forecast flag, so the last quarter with a whole-number CPI index is treated as the last actual. The fiscal file carries its own flag, which is used directly.

## What is not built yet

This was built in under a day as a first release. The following are planned and tracked in the repository, and are not claimed as done:

| Item            | Status                                                                                                                                                             |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Failure alert   | The pipeline is scheduled, but nothing emails or messages anyone when a run fails; a failure is visible in the Fabric Monitor hub. Next step is an alert activity. |
| Dev, Test, Prod | One workspace so far. Next step is Fabric Git integration and a deployment pipeline with an acceptance checklist before release.                                   |
| Reconciliation  | Tests check structure and integrity. A check of selected figures against the Treasury's published tables is still to add.                                          |
| Actuals         | Stats NZ outturns (GDP, CPI, unemployment) as a second source, to show actual against forecast.                                                                    |
| Report polish   | No reusable theme file yet; the publication dates in the report footer are typed, not read from the data.                                                          |

## How it was built

I set the requirements (one cloud platform, automated ingestion from a predictable public source, change detection, an architecture that can take further sources, and a full release life cycle rather than a one-off build) and chose the subject and approach. I directed an AI assistant (Claude), which wrote the notebook code and the DAX measures, built the pipeline, and carried out much of the Fabric configuration and report build to my direction, and I reviewed the results. The decision log and status file in the repository record what was decided and why.

## Platform

Microsoft Fabric (lakehouse, PySpark notebooks, Data Factory pipeline, Direct Lake and import semantic models), Power BI service, GitHub. Built entirely in the browser. Data: the Treasury, Economic and Fiscal Update data files ([budget.govt.nz](https://budget.govt.nz)).

## The report

The two pages that follow are exported from the live Power BI report.

# Treasury Forecast Tracker

Loaded by a scheduled Microsoft Fabric pipeline (runs daily at 7:00 am)

Independent demonstration by Brett McDonald; not a Treasury product

Data logs

### Measure

- ☐ Real GDP growth
- ☐ CPI inflation
- ☐ Unemployment rate
- ☐ OBEGAL (excluding ACC)
- ☐ OBEGAL (excluding ACC), % of GDP
- ☒ Net core Crown debt, % of GDP
- ☐ Core Crown expenses, % of GDP
- ☐ Core Crown tax revenue, % of GDP

### Purpose and function

**What it does:** Compares the Treasury's three most recent forecast rounds across eight headline economic and fiscal measures.

**Why:** To show how the outlook was revised from one round to the next, without opening three sets of tables.

**How to use it:** Select a measure above. The chart traces each round, the note above it states what changed in the latest round, and the table gives the figures. Data logs shows every pipeline run and the checks the data passed.

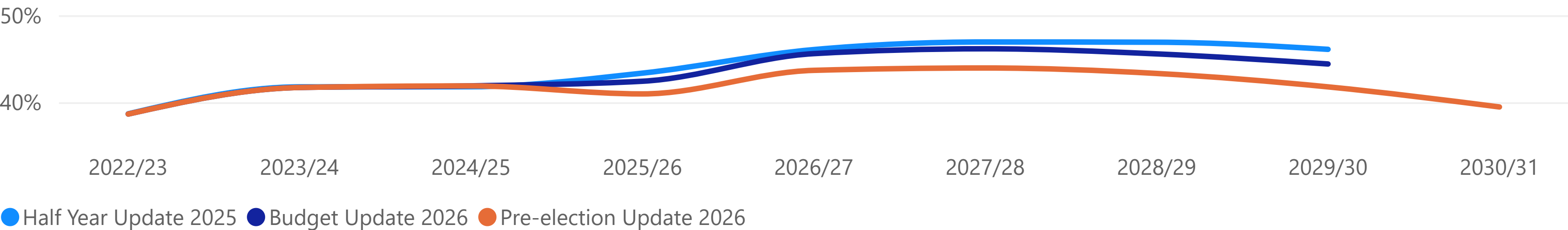
**Source data published:** Half Year Update 2025 on 16 Dec 2025; Budget Update 2026 on 28 May 2026; Pre-election Update 2026 on 29 Sep 2026.

### What changed in the latest update (calculated automatically from the loaded data)

Net core Crown debt, % of GDP: the Pre-election Update 2026 revised 4 of the 4 forecast periods it shares with the Budget Update 2026 (0 up, 4 down). The largest change is for 2029/30: down 2.6 percentage points (from 44.4 to 41.8).

### Net core Crown debt, % of GDP by forecast round

Unit: % of GDP



### Figures by forecast round (% of GDP)

| Period  | Half Year Update 2025 | Budget Update 2026 | Pre-election Update 2026 |
|---------|-----------------------|--------------------|--------------------------|
| 2022/23 | 38.7%                 | 38.6%              | 38.6%                    |
| 2023/24 | 41.8%                 | 41.7%              | 41.7%                    |
| 2024/25 | 41.8%                 | 41.9%              | 41.9%                    |
| 2025/26 | 43.3%                 | 42.4%              | 41.0%                    |

**Next publishing date:** not yet set. The Treasury has said the timing of its next forecast update will be determined later.

Source: the Treasury, Economic and Fiscal Update data files (budget.govt.nz)

Created by Brett McDonald on 8 October 2026

# Pipeline and data quality

Independent demonstration by Brett McDonald; not a Treasury product

What each run of the Fabric pipeline found, and the data checks passed at the last load

Return to tracker

Ingestion log: every run, newest first (unchanged = same file as last time; 404 = no file published for that round)

| Run time (UTC)      | Round   | Status        | HTTP status | File size (bytes) | Source file                                                                        |
|---------------------|---------|---------------|-------------|-------------------|------------------------------------------------------------------------------------|
| 2026-10-08 00:54:11 | BEFU26  | unchanged     | 200         | 48,676            | https://budget.govt.nz/budget/excel/befu2026/befu26-economic-forecasts-data.xlsx   |
| 2026-10-08 00:54:11 | HYEFU25 | unchanged     | 200         | 110,564           | https://budget.govt.nz/budget/excel/hyefu2025/hyefu25-economic-forecasts-data.xlsx |
| 2026-10-08 00:54:11 | HYEFU26 | not_published | 404         |                   | https://budget.govt.nz/budget/excel/hyefu2026/hyefu26-economic-forecasts-data.xlsx |
| 2026-10-08 00:54:11 | PREFU26 | unchanged     | 200         | 60,712            | https://budget.govt.nz/budget/excel/prefu2026/prefu26-economic-forecasts-data.xlsx |
| 2026-10-08 00:50:16 | BEFU26  | unchanged     | 200         | 48,676            | https://budget.govt.nz/budget/excel/befu2026/befu26-economic-forecasts-data.xlsx   |

Automated data checks: all must pass before the reporting tables refresh

| Check                                         | Passed | Detail                                                                              |
|-----------------------------------------------|--------|-------------------------------------------------------------------------------------|
| every fact series is in dim_series            | True   |                                                                                     |
| fact has rows                                 | True   | 12411                                                                               |
| forecast flag set on every row                | True   |                                                                                     |
| headline source series present in every round | True   | ['lhurzq', 'ngdpp_zq', 'nzgpf0024', 'nzgpf0begalexacc', 'nznaac0317_fisc', 'pcpiq'] |
| no duplicate fact keys                        | True   |                                                                                     |
| no null values in fact                        | True   |                                                                                     |
| quarterly GDP series has no gaps              | True   | 0                                                                                   |